

100 Days of Azure DevOps

Day 49 Handout — Plan in CI on a clean agent. Apply the plan you reviewed — not a second plan nobody saw

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — plan artifact, then a gated apply

Stage	Command	Output
CI Plan	terraform plan -out=tfplan (or az deployment group what-if)	Reviewed diff + binary/json plan file
Publish	PublishPipelineArtifact@1 artifact: tfplan	The exact plan Apply must consume
Apply	terraform apply tfplan after approval / apply=true	Mutates Azure. Same plan, not a fresh one
Bicep track	what-if in Plan; deployment create in Apply	Same brake. No 'it worked on my laptop'
Identity	azure-100-sc scoped to the lab RG	Not Owner on the subscription

Architecture B — brakes (pick at least one)

Brake	YAML / UI	Use
Environment approval	deployment job environment: prod (Day 38)	Human click before apply
Variable gate	condition: and(succeeded(), eq(variables['apply'], 'true'))	Manual run with apply=true
-auto-approve on apply	Only after the gate, on the saved plan	Never on a fresh unreviewed plan to prod
Laptop apply	Homework	Not how grown-ups change shared env

Architecture C — pitfalls that re-plan in the dark

Pitfall	What you see	Fix
apply without -out	Second plan at apply time; drift snuck in	apply the tfplan file from the artifact
Plan on laptop, apply in CI	Different env vars / different plugin versions	Both on the same agent image + versions
Owner SPN	Pipeline can delete the subscription	Scope the service connection to rg-day49-lab
Auto-approve prod	YOLO with extra YAML	Approval or apply=true. Not hope

One-liner to remember

IaC without a pipeline is homework; IaC in a pipeline is how grown-ups change prod — apply the plan you reviewed.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org + personal subscription. Hosted ubuntu-latest. Brake required. Destroy after.

- 1 Pipeline stage Plan: terraform init -input=false, terraform plan -out=tfplan (Bicep: what-if). Publish tfplan (or the what-if log).
- 2 Stage Apply dependsOn Plan, condition succeeded(), and either environment: prod approval or eq(variables['apply'], 'true').
- 3 Apply step: terraform apply -input=false tfplan (download the artifact first). Do not run a new plan there.
- 4 Queue once with apply=false / no approval path — confirm Apply skipped. Queue again with the brake released.
- 5 If the apply changes something the published plan did not show, stop. Versions or backend drifted.
- 6 Destroy via pipeline or local after the screenshot. Do not leave rg-day49-lab.

Plan artifact + gated apply (Terraform track)

```
# pipelines/iac.yml (hosted ubuntu-latest)
stages:
- stage: Plan
  jobs:
  - job: PlanJob
    steps:
    - script: |
        terraform init -input=false
        terraform plan -out=tfplan
        workingDirectory: infra/tf
    - publish: infra/tf/tfplan
      artifact: tfplan
- stage: Apply
  dependsOn: Plan
  condition: and(succeeded(), eq(variables['apply'], 'true'))
  jobs:
  - deployment: ApplyJob
    environment: prod
    strategy:
      runOnce:
        deploy:
          steps:
          - download: current
            artifact: tfplan
          - script: terraform apply -input=false $(Pipeline.Workspace)/tfplan/tfplan
            workingDirectory: infra/tf
```

Done checklist

- Plan ran on the agent, not only on the laptop
- Apply consumed the published plan file
- A skipped Apply run exists (brake worked)
- Service connection is not subscription Owner

Tomorrow — Day 50

Phase 5 mini project — one RG + storage or webapp skeleton from the chosen IaC in a pipeline, then destroy.